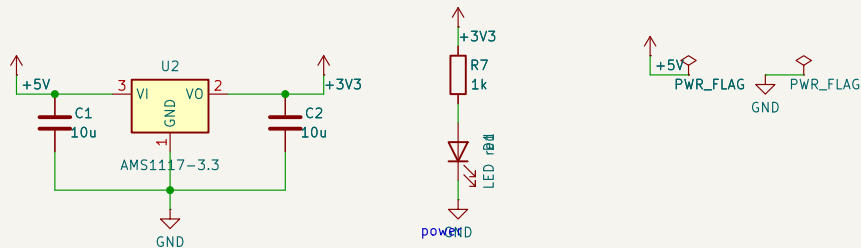
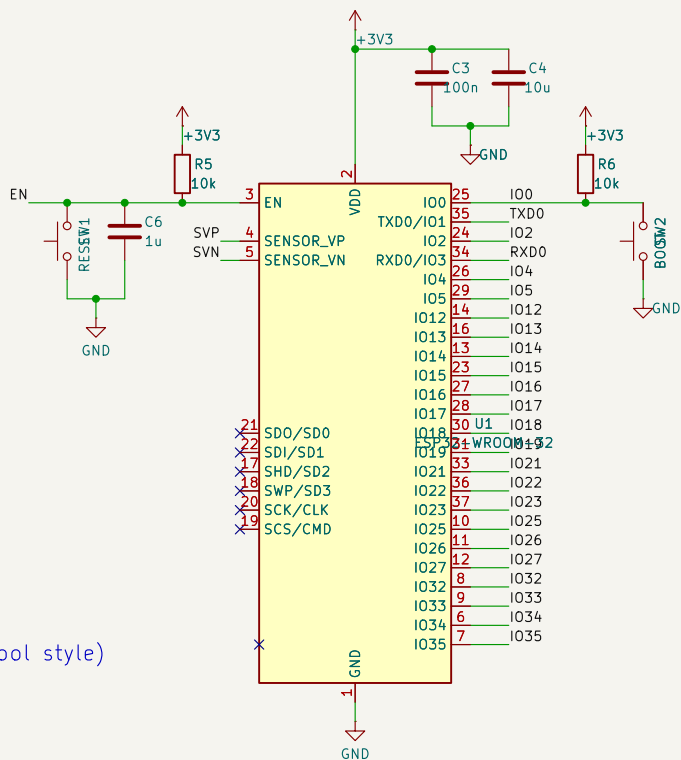


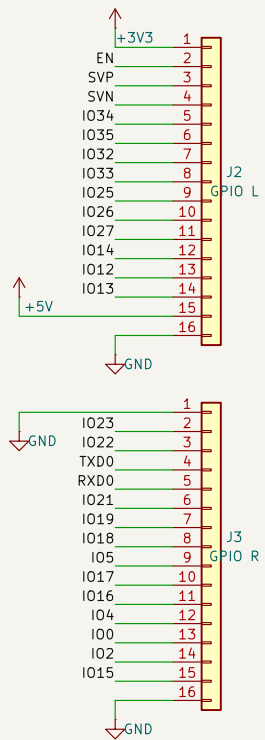
3.3 V regulator (USB 5 V -> AMS1117-3.3 -> 3V3)



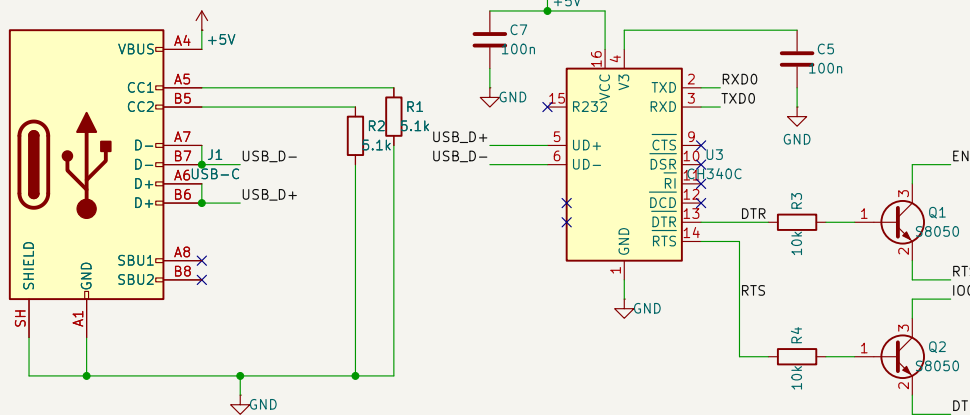
ESP32-WROOM-32 (EN/BOOT buttons, 3V3 decoupling)



GPIO breakout (2 x 16, module pin order)

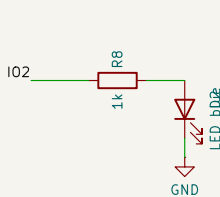


USB-UART bridge (CH340C, no crystal)
USB-C input (5 V + USB 2.0 data, CC pull-downs for C-to-C cables)



Auto-reset (DTR/RTS -> EN/I00, esptool style)

User LED (IO2)



I2C header (IO21 SDA / IO22 SCL, 4k7 puW headers (DS18B20 pinout: GND, DQ, 3V3)

